

# Comparative analysis of different methods for scoring social contact seeking in adolescent mice

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Social interactions and their variability are important measures in neuroscientific research.

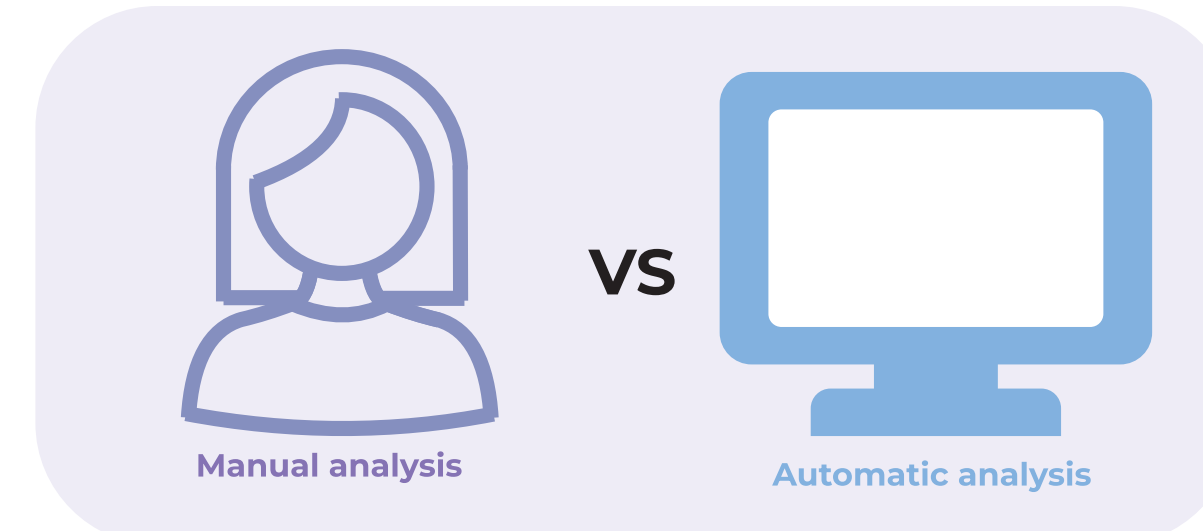
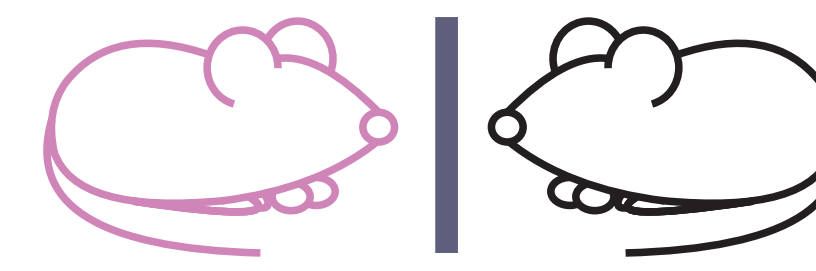
Because detecting such subtle, reward-specific changes critically depends on how social interactions are quantified, the reliability of behavioral scoring becomes essential. Automatic analysis of recorded mouse behavior greatly accelerates the research and removes the risk of bias.

However, is it as accurate as scored by a human observer?

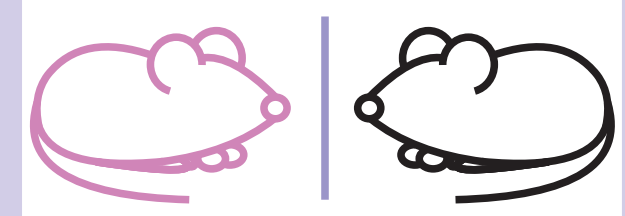
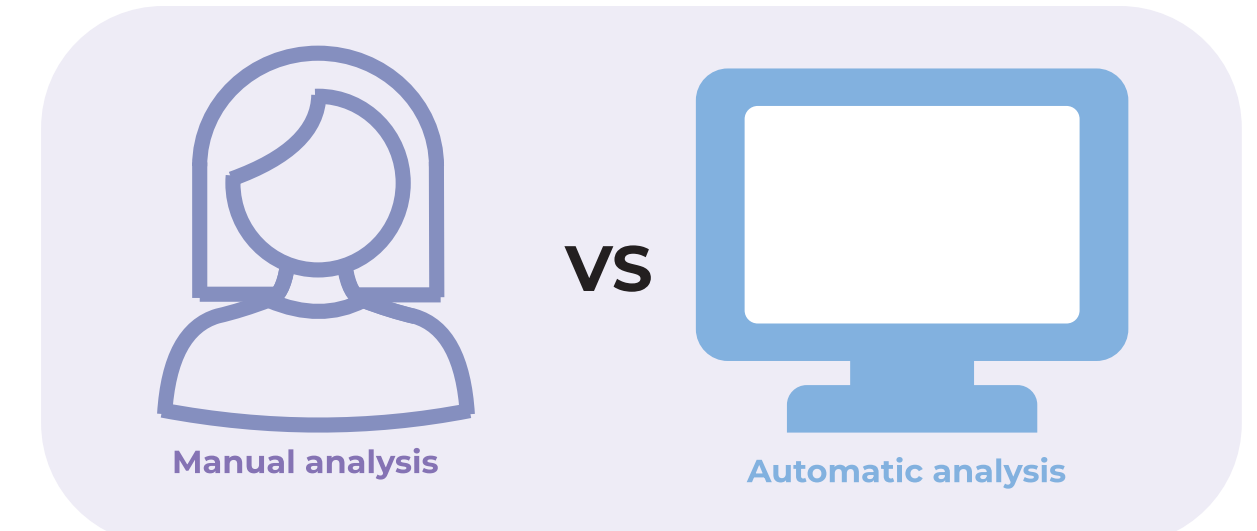
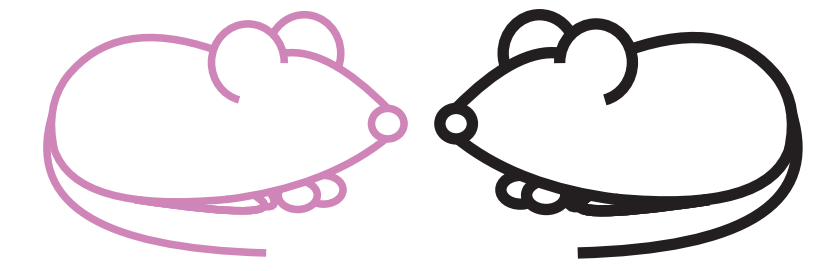
To verify whether an automated analysis program gives results comparable to human observer in

measuring social interactions I compared results obtained with manual (BORIS) and automatic (EthoVision v.15) scoring in two social behavioral paradigms- with and without a divider between two mice.

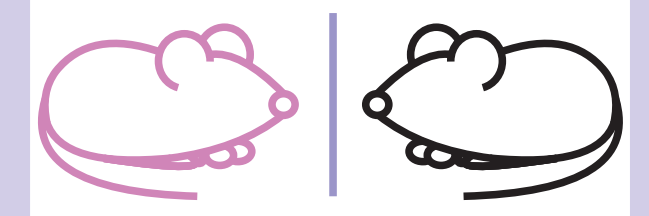
## Non-direct social interaction



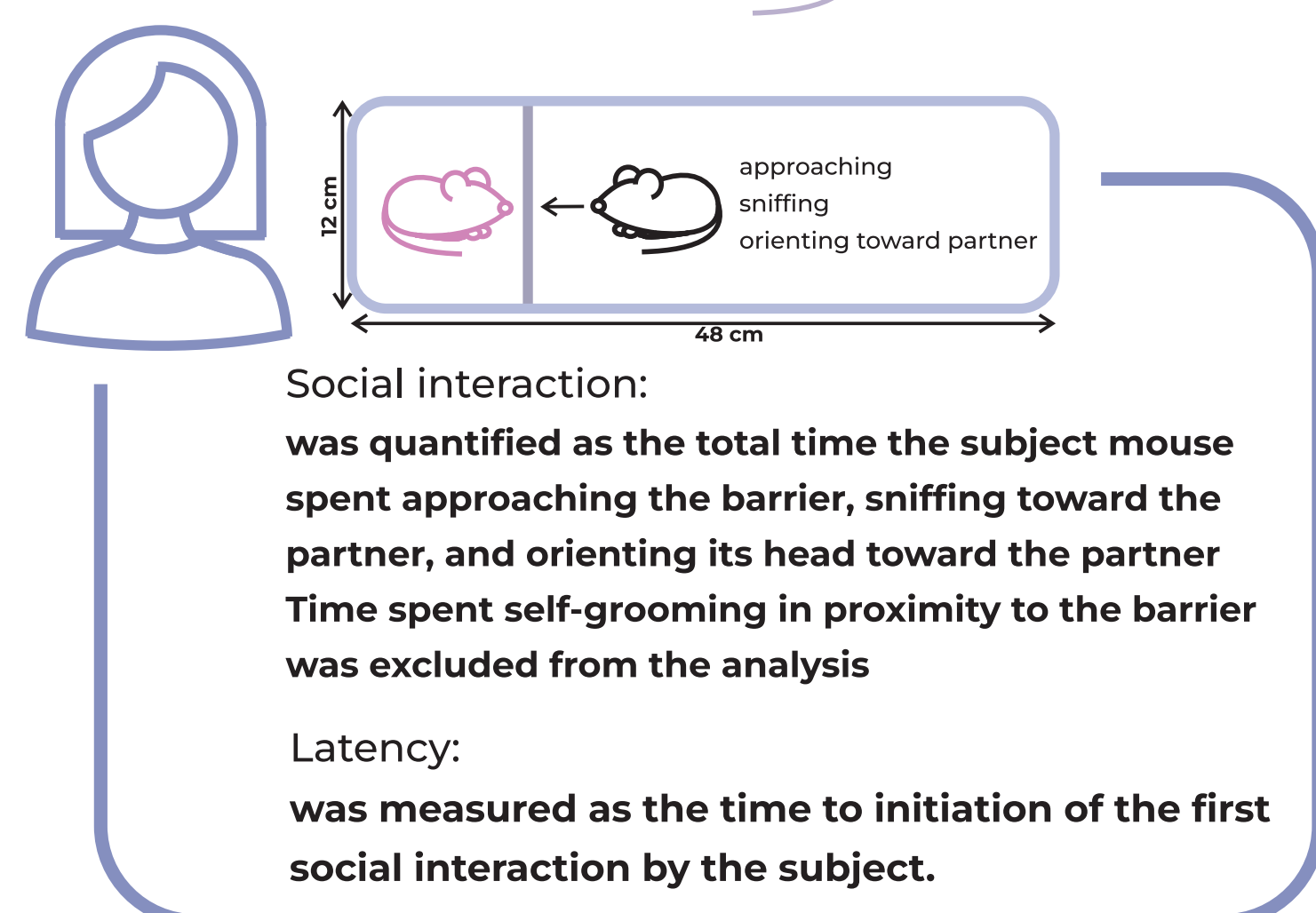
## Direct social interaction



## When there is a barrier- both methods capture the same effects

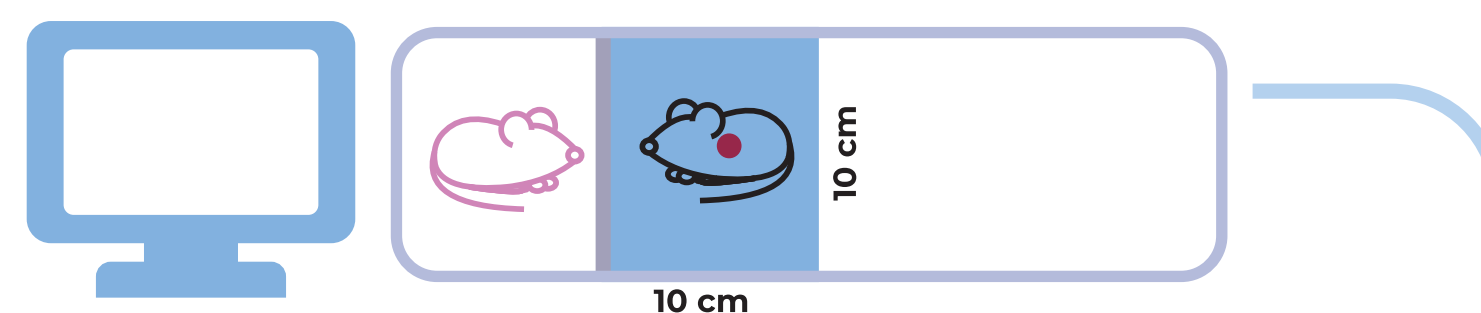


33 days old x 10  
38 days old x 18  
43 days old x 10



Social interaction: was quantified as the total time the subject mouse spent approaching the barrier, sniffing toward the partner, and orienting its head toward the partner. Time spent self-grooming in proximity to the barrier was excluded from the analysis.

Latency: was measured as the time to initiation of the first social interaction by the subject.

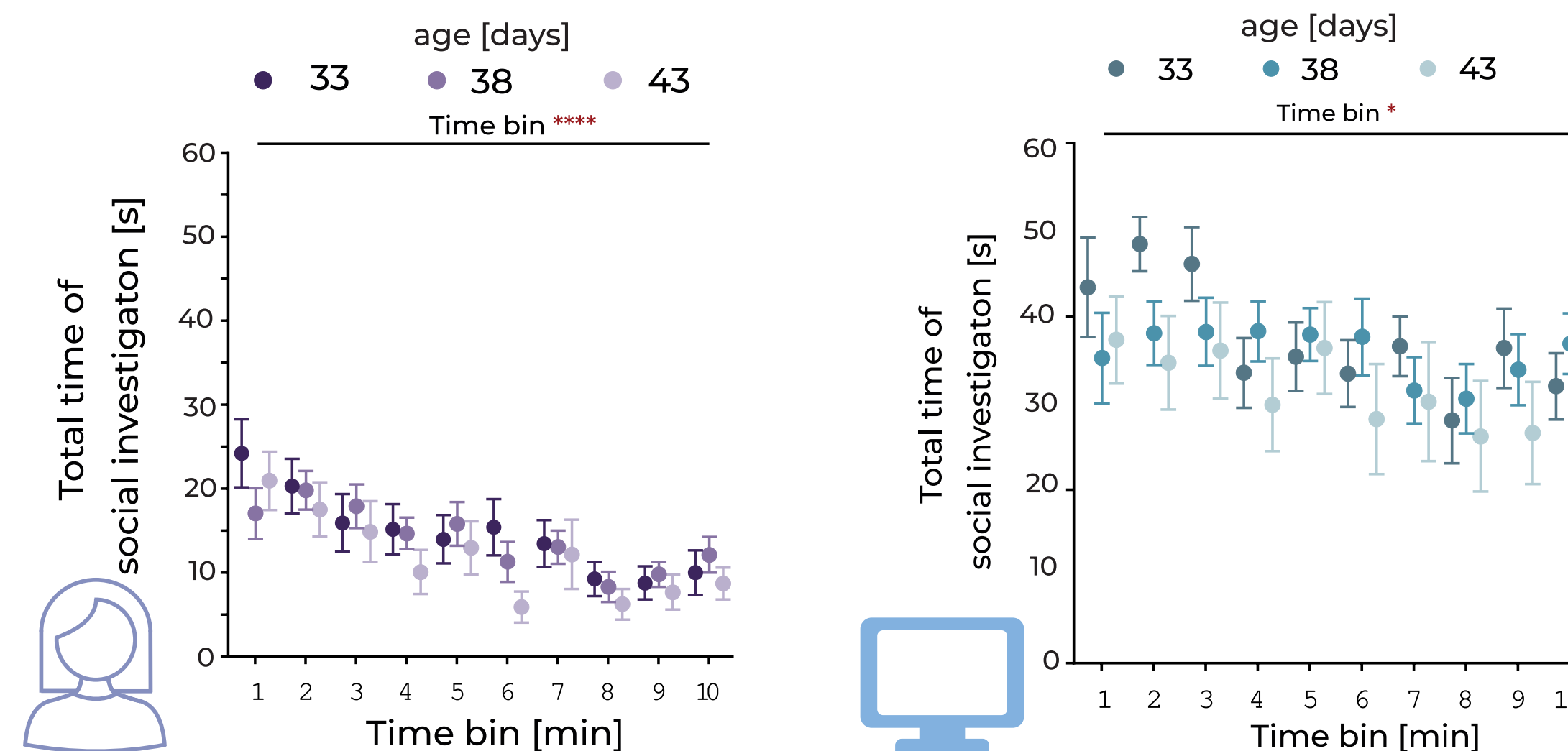


Social interaction: was quantified as the total time the center point of a subject mouse was detected in predefined 10x10 cm zone.

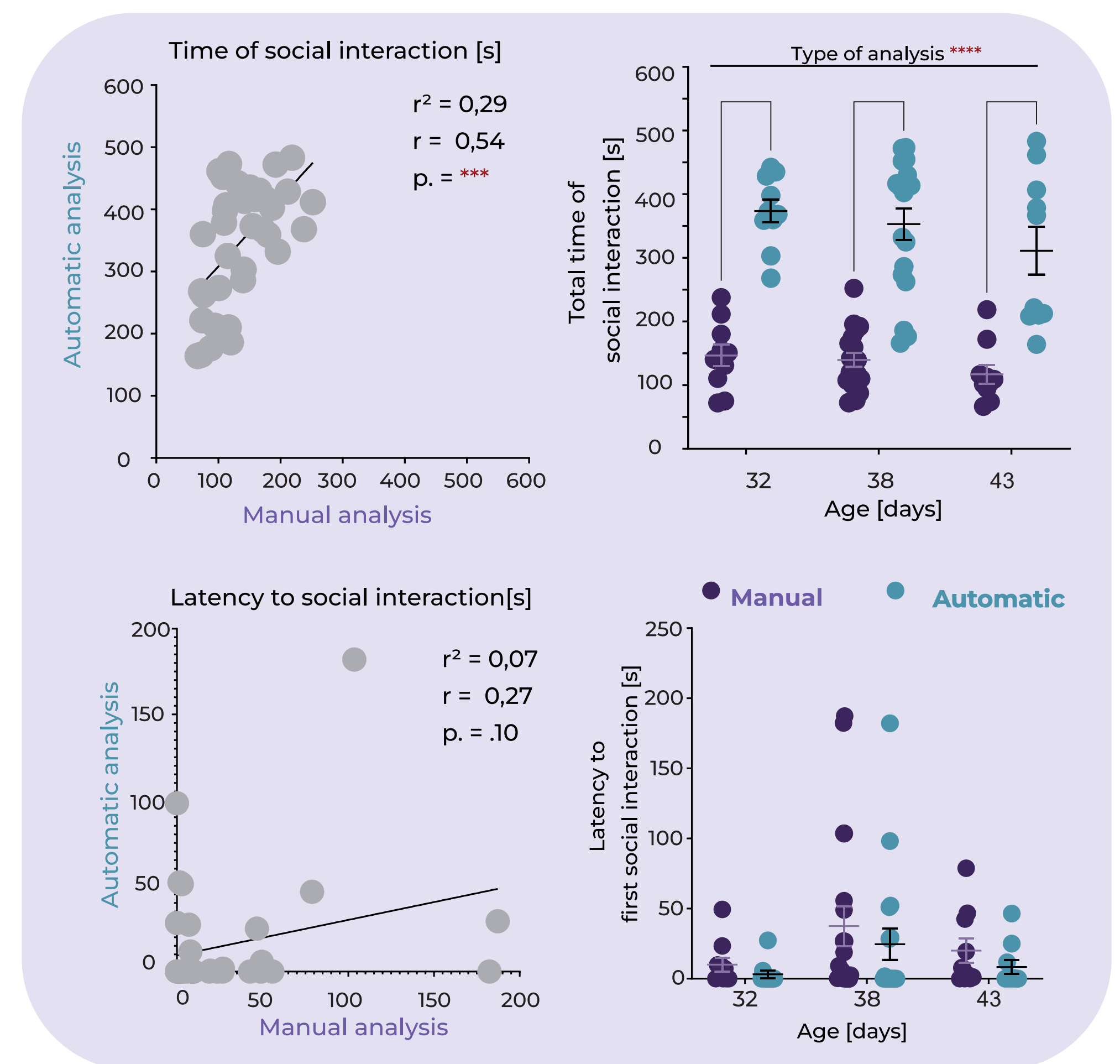
Latency: was measured as the time to the first entry of the subject's center point into the predefined zone.

## Does motivation for social interaction vary depending on the stage of adolescence?

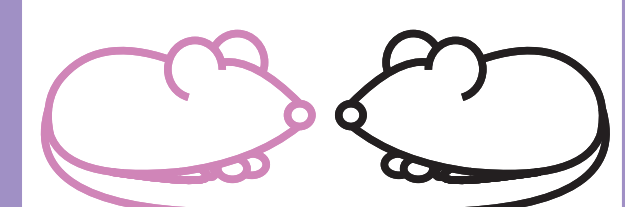
Analysis of social investigation time in 1-min intervals was performed across all age groups. Interaction time decreased over the course of 10 min regardless of age, and this pattern was captured by both manual, and automated analysis. Based on this in further comparison I used cumulative time of interaction obtained via both methods.



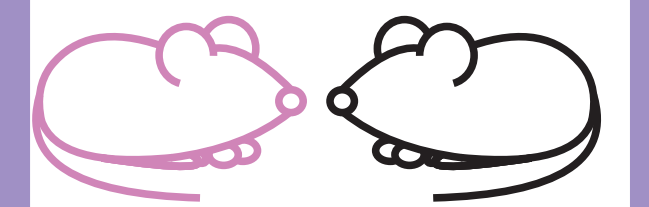
To assess whether automatic analysis captured the same effect as manual scoring total interaction time- the primary measure of social motivation- was evaluated. Results obtained via both methods showed positive correlation, and same result- no effect of age on time on total interaction- even though automatic analysis consistently yielded higher results. In addition latency to first interaction was compared, and again results were consistent across scoring



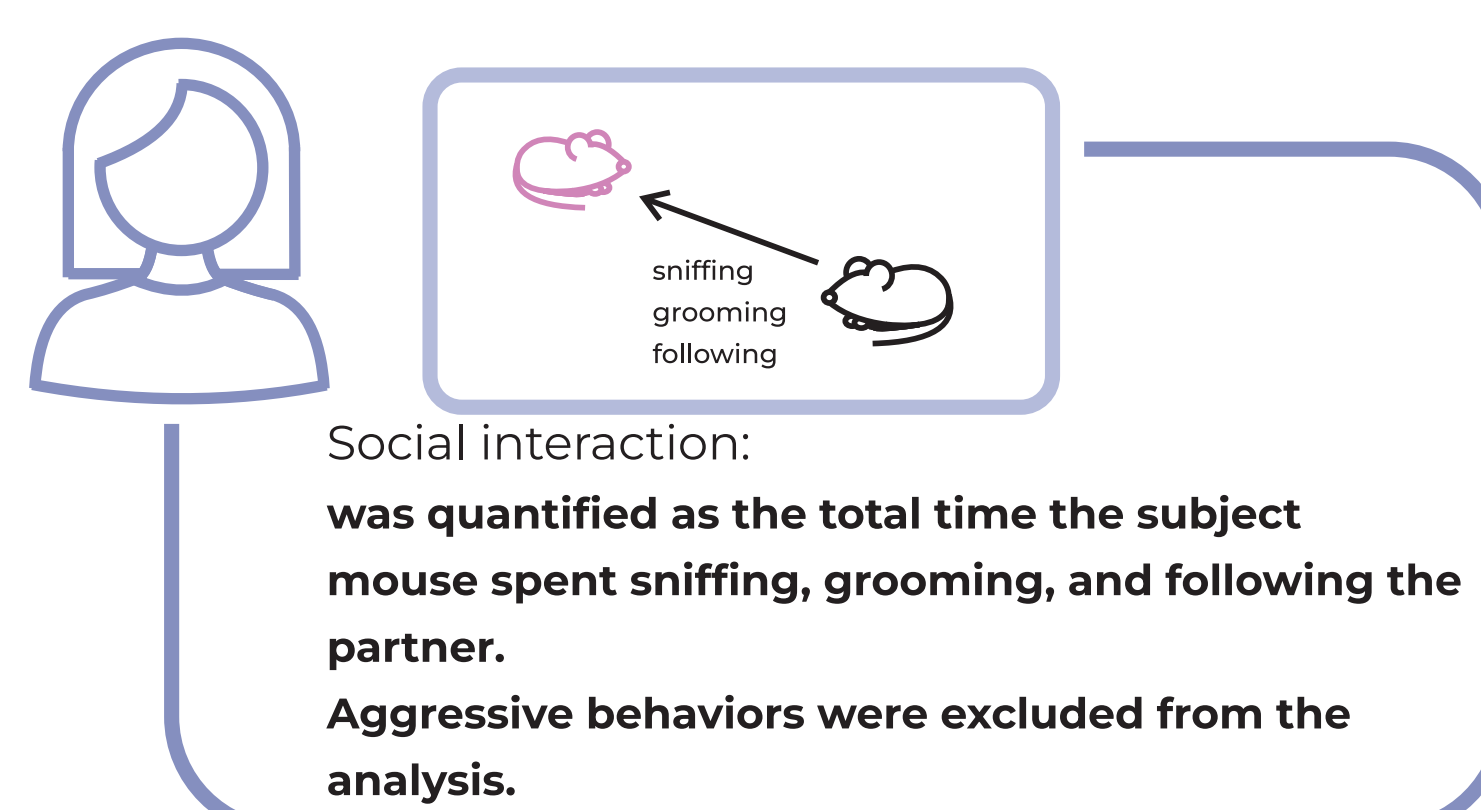
methods- there was no significant difference between age groups. Additionally this parameter did not correlate with interaction time.



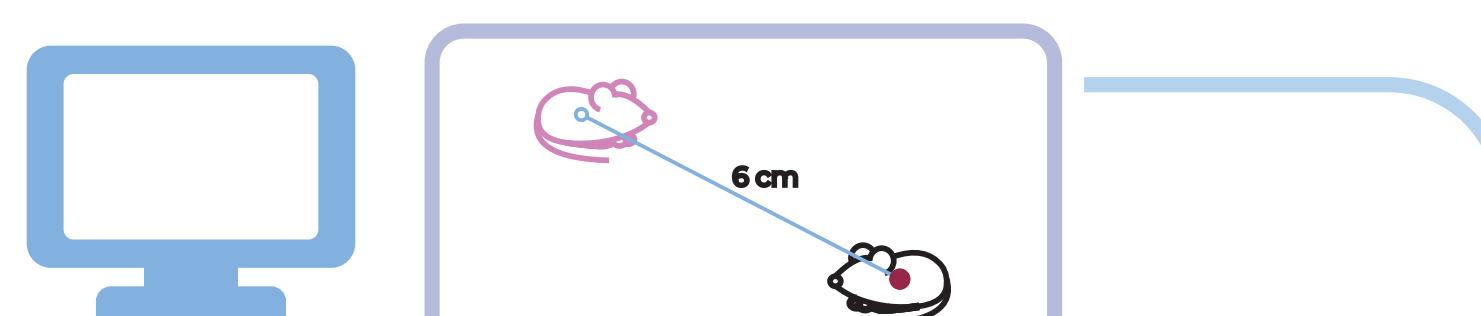
## Automated scoring becomes less reliable with direct interaction



control x 3  
isolation x 4



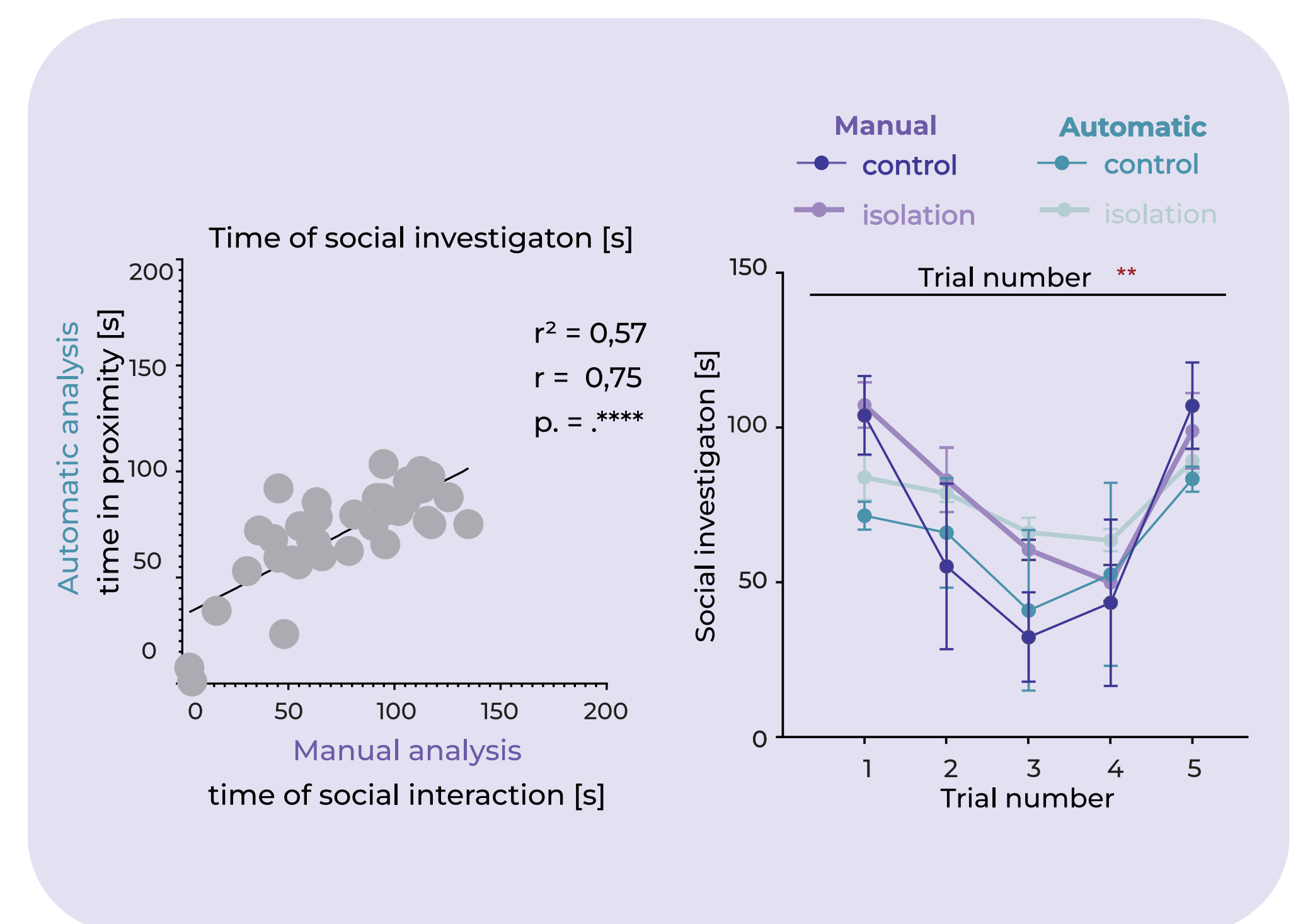
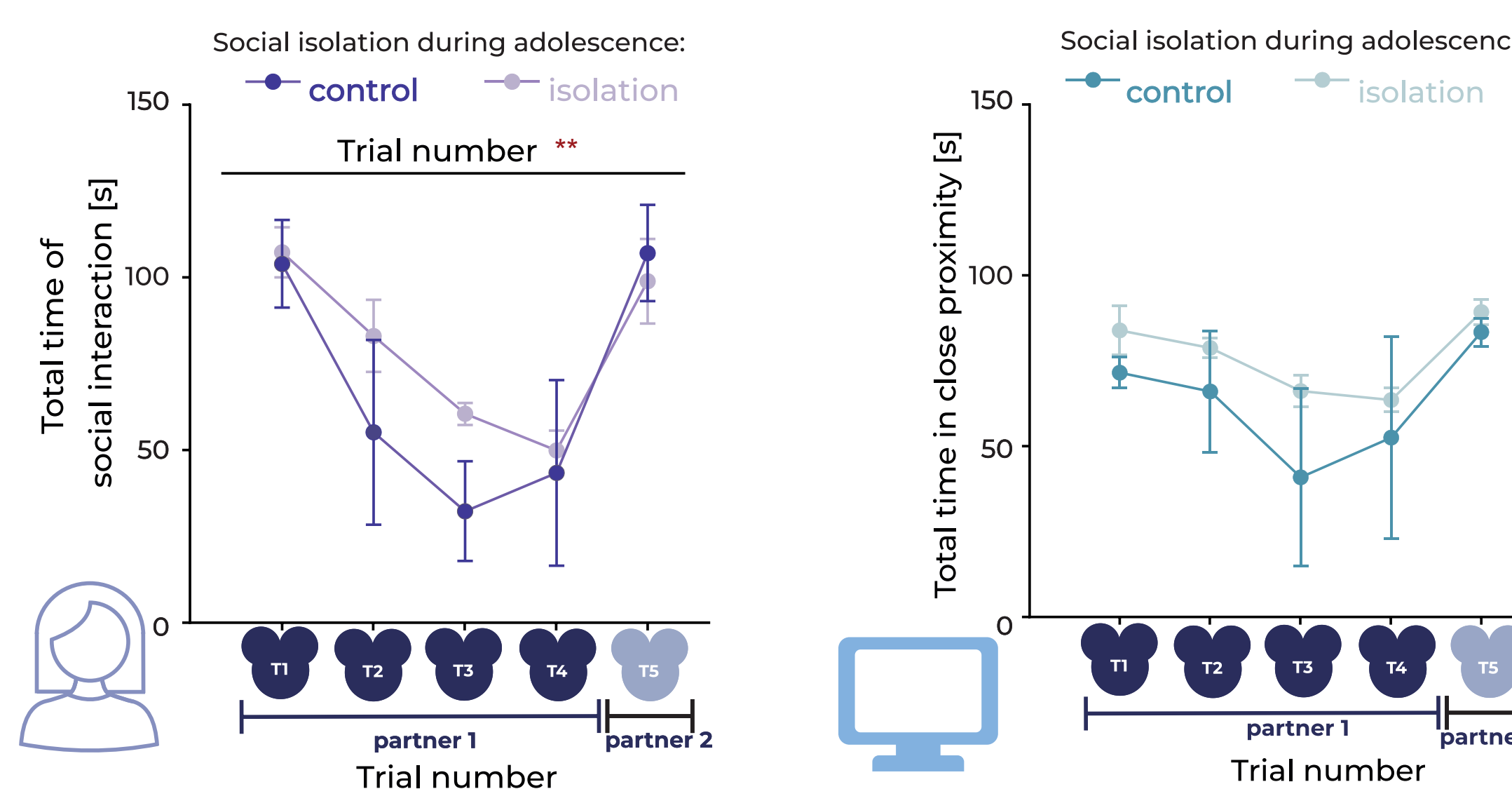
Social interaction: was quantified as the total time the subject mouse spent sniffing, grooming, and following the partner. Aggressive behaviors were excluded from the analysis.



Social interaction: was quantified as the cumulative time during which the center points of the subject and partner mice were in proximity (<6 cm).

## Does isolation during adolescence affect time spent on social interaction?

To assess how social memory is affected by isolation, a social recognition test was performed. Manual analysis showed, that there is no significant effect of isolation on total social investigation time, and the latter depended solely on the trial number. In comparison, automated analysis failed to capture such difference. Further evaluation between all subjects confirmed, that social investigation time depended solely on trial number, with most significant difference between trial no. 1 & 4.

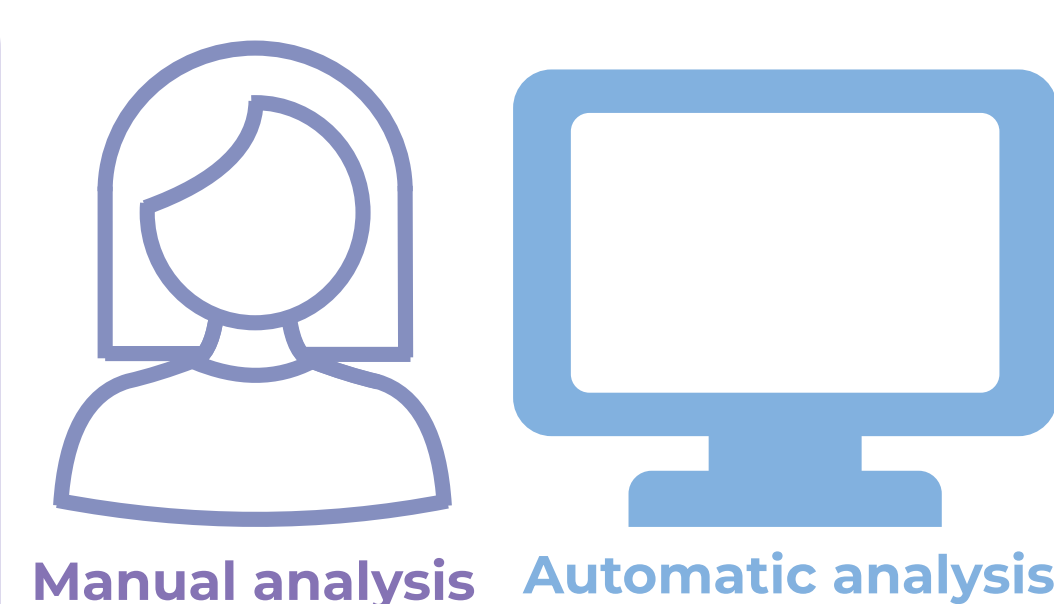


Results obtained via both methods were positively correlated. The program's inability to capture differences demonstrates that it cannot be a reliable tool for this task, due to its failure to capture the most fundamental difference in this task.

## Conclusions

- Manual and automated scoring are **positively correlated** indicating that automated analysis captures general social behavior trends
- Automated and manual methods are **not directly interchangeable**, as automated scoring tends to systematically over- or underestimate social interaction time
- Automated scoring is **reliable** in **simple, spatially constrained** social paradigms, as it leads to the same conclusions as manual scoring
- Positive correlation between methods **does not ensure equivalent sensitivity or effect size**, as automated scoring fails to detect meaningful behavioral effects in complex, unconstrained social interaction test

- More accurate estimation of absolute values
- Detects significant effects regardless of paradigm
- Highly sensitive
- Time consuming
- Carries subjective bias



- Carries systematic bias (over/under- estimation) of absolute values
- In simple paradigm (with barrier) detects same effects as an observer (= effective!)
- Fails to detect effects in more precise paradigms (direct contact between subjects)
- Reduced sensitivity in certain conditions



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